

## LESSON 7-6 • SESSION 1 • PRACTICE • FORM B

## Distance Between Points on a Grid

Learning target – I can find the distance between two points that share a coordinate. • 6.NS.C.8

## REMEMBER

- Same **y-coordinate**: the points are side to side. Use the x-coordinates.
- Same **x-coordinate**: the points are up and down. Use the y-coordinates.
- **Opposite signs**: add the absolute values. **Same sign**: subtract them.  $(5, -2)$  to  $(5, -9)$  is  $9 - 2 = 7$ .

## WORKED EXAMPLE

Each unit is 1 meter. Find the distance from  $P(-9, 4)$  to  $Q(3, 4)$ , and from  $P$  to  $R(-9, 11)$ .

**Step 1**  $P$  and  $Q$  share  $y = 4$ , so use  $x$ .  $-9$  and  $3$  have opposite signs:  $9 + 3 = 12$  m.

**Step 2**  $P$  and  $R$  share  $x = -9$ , so use  $y$ .  $4$  and  $11$  are both positive:  $11 - 4 = 7$  m.

**Answer**  $PQ = 12$  m and  $PR = 7$  m

**WORDS** distance (*distancia*) • absolute value (*valor absoluto*) • coordinate plane (*plano de coordenadas*)

## PRACTICE

show your work • label units

1. Find the distance between each pair of points.

a.  $(2, 12)$  and  $(2, 5)$  \_\_\_\_\_

b.  $(-9, 3)$  and  $(7, 3)$  \_\_\_\_\_

c.  $(-12, 8)$  and  $(-12, -5)$  \_\_\_\_\_

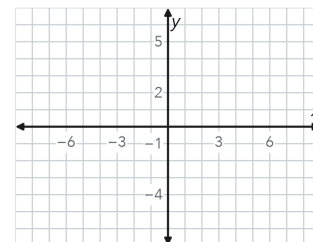
d.  $(-5, 1)$  and  $(9, 1)$  \_\_\_\_\_

2. **Town Map Walk.** Each unit is 1 kilometer. Market  $(-6, 4)$ , Museum  $(8, 4)$ , Station  $(-6, -7)$ .

a. Plot and label the three places.

b. Market to Museum \_\_\_\_\_ km

c. Market to Station \_\_\_\_\_ km



3. **Drone Flight.** A drone flies straight from  $(-6, 11)$  to  $(-6, -5)$ . Each unit is 1 meter. How far does it fly?

**Hint** Same  $x$ , so use  $y$ .  $11$  and  $-5$  have opposite signs.

Show your work.

**Answer** \_\_\_\_\_ m

4. **One Quadrant.** Find the distance from  $(-15, -8)$  to  $(-15, -20)$ .

**Hint** Both  $y$ -values are negative, so subtract.

Show your work.

**Answer** \_\_\_\_\_ units

5. **Missing Point.** Point  $G$  is at  $(9, -2)$ . Point  $H$  is 14 units directly to the left of  $G$ . Name point  $H$ .

**Hint** Moving left changes only  $x$ :  $9 - 14$ .

Show your work.

**H** = \_\_\_\_\_

6. **Window Washer.** A window washer moves from  $(19, 17)$  to  $(19, 8)$ . Each unit is 1 foot. How far does the washer move?

**Hint** Are  $17$  and  $8$  on the same side of  $0$ ?

Show your work.

**Answer** \_\_\_\_\_ ft

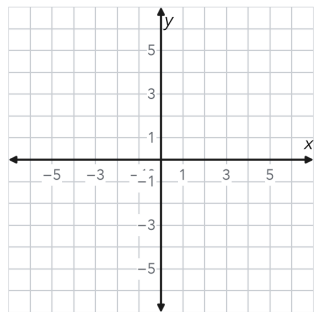
- 7. Bus Route.** Omar rides a bus straight from  $(-15, -3)$  to  $(13, -3)$ . Each unit is 1 block. How far does he ride?

Show your work.

**Answer** \_\_\_\_\_ blocks

- 8. City Grid.** Each unit is 1 block. Daniel walks from a library at  $(-5, 4)$  to a school at  $(6, 4)$ , then to a park at  $(6, -2)$ . Plot the walk on the grid.

- a. Library to school \_\_\_\_\_ blocks  
 b. School to park \_\_\_\_\_ blocks  
 c. Total walk \_\_\_\_\_ blocks



### FIND THE ERROR

explain the mistake, then fix it

- 9. Amina's Work.** To find the distance from  $(3, -8)$  to  $(3, 5)$ , Amina wrote  $8 - 5 = 3$  units. Explain her mistake. Then find the correct distance.

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### TEST PRACTICE

choose the best answer

- 10.** The distance from  $(6, 1)$  to  $(6, 10)$  is –  
**A** 4 units      **B** 9 units      **C** 11 units      **D** –9 units
- 11.** Two points with the same x-coordinate lie on a –  
**A** horizontal line      **B** vertical line      **C** diagonal line      **D** circle

### SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

|                       |                      |                      |                       |                       |                       |                       |                             |                             |                             |
|-----------------------|----------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------------|-----------------------------|-----------------------------|
| <b>28</b><br><b>I</b> | <b>9</b><br><b>G</b> | <b>2</b><br><b>X</b> | <b>25</b><br><b>J</b> | <b>17</b><br><b>N</b> | <b>16</b><br><b>O</b> | <b>12</b><br><b>R</b> | <b>(23, -2)</b><br><b>Z</b> | <b>(9, -16)</b><br><b>Q</b> | <b>(-5, -2)</b><br><b>I</b> |
|                       |                      | #3                   | #4                    | #5                    | #6                    | #7                    | #8c                         |                             |                             |

### CHECK YOURSELF

☐ I can do this on my own    ☐ I need a little help    ☐ I have a question about #\_\_\_\_\_